SquiDBase: A Centralized Community Resource of Microbial Paired Squiggle-Sequence Data

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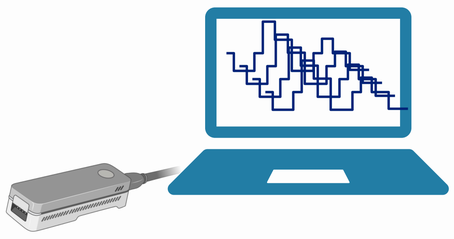
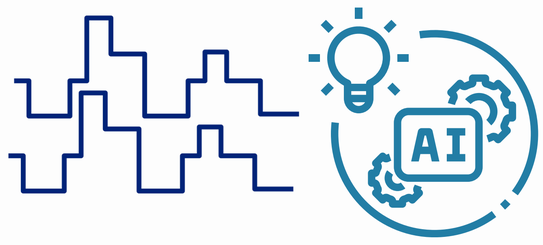
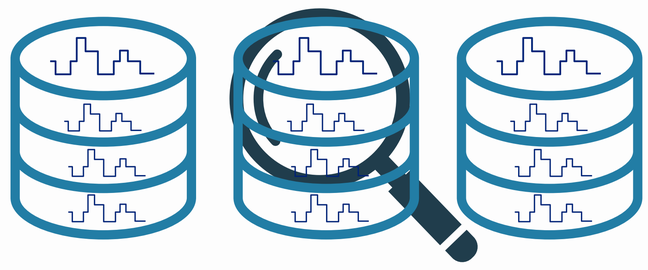
Challenges in handling and storing raw nanopore data



Raw nanopore data ('squiggles') are voluminous and challenging to manage effectively.

Enabling **computational dataset generation for AI purposes** can unlock the rich potential of squiggle data.



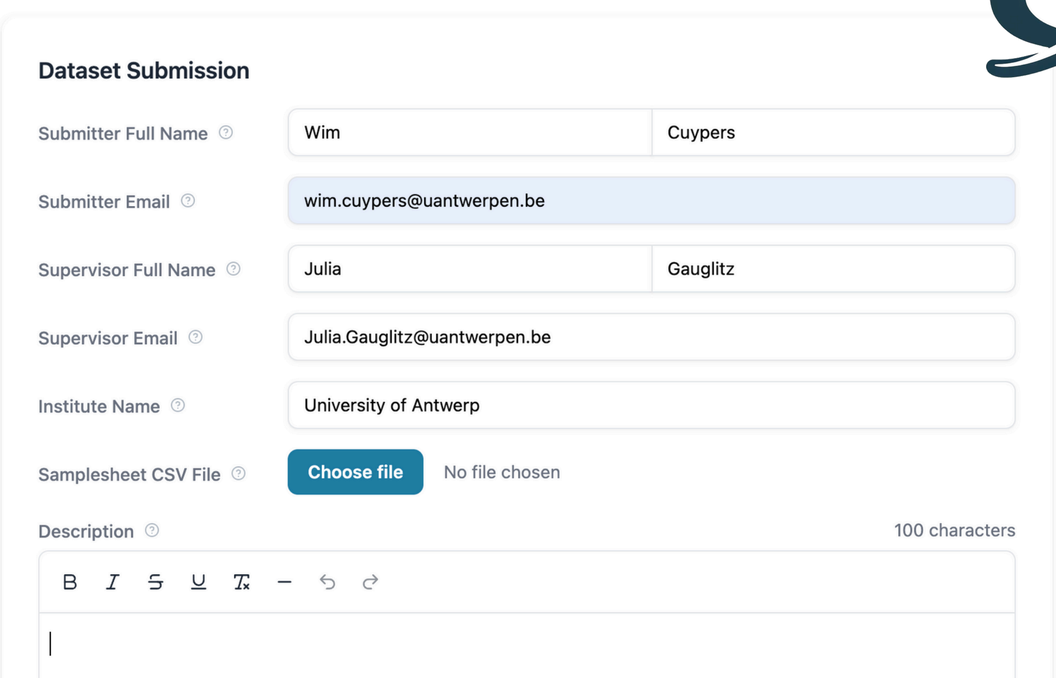


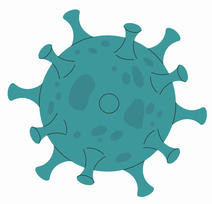
Aggregating specific subsets of squiggle data for developing AI algorithms is challenging due to their **distribution across various databases**.

SquiDBase enables data collection and query-driven subset generation for AI

SquiDBase features

- User-friendly interface facilitating data upload, download, and queries.
- Support for raw POD5 file uploads and structured metadata for enhanced reusability.
- Easy to acccess data from # submissions.



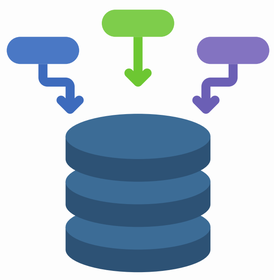


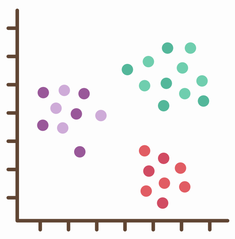
Genomic data from pathogens in human blood (*Plasmodium*) and cultured viral isolates (SARS-CoV-2 variants, Dengue virus, Monkeypox virus).

Roadmap

Automated removal of human genetic information via a NextFlow pipeline.

Open invitation for the research **community** to **contribute and expand this vital resource**.





Ultrafast squiggle matching, *e.g.* for ultrafast pathogen detection in metagenomic samples.

Read more & contact

squidbase.org